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Photosynthesis converts light energy into chemical energy stored in glucose.

The overall equation for photosynthesis is: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light produces } \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

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Chlorophyll is the primary pigment that absorbs light, mainly in the red and blue wavelengths.

The light-dependent reactions occur in the thylakoid membranes of the chloroplast.

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The Calvin cycle (light-independent reactions) takes place in the stroma.

ATP and NADPH produced in the light reactions power the Calvin cycle.

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Carbon fixation is the process by which CO_2 is incorporated into organic molecules via RuBisCO.

The rate of photosynthesis increases with light intensity up to a saturation point.

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Photorespiration occurs when RuBisCO fixes O₂ instead of CO₂, reducing photosynthetic efficiency.

C₄ plants like maize use a two-stage carbon fixation to minimise photorespiration.

CAM plants open stomata at night to reduce water loss during carbon fixation.

Oxygen produced during photosynthesis comes from the splitting of water molecules.